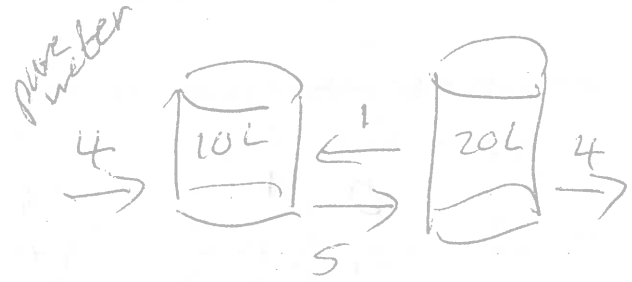


W

- Sketch a possible tank system.

Compute the two exponential decay rates of the system

values of $A!!$

$$\lambda_1 = -0.58$$
$$\lambda_2 = -0.17$$

slower more important

1. The ODE for a tank system is $y' = Ay$ with $A = \begin{pmatrix} -\frac{5}{20} & \frac{2}{10} \\ \frac{5}{20} & -\frac{5}{10} \end{pmatrix}$.

Sketch a possible tank system.

Compute the exponential decay rates of the system

ODE

$$Ri + q/C + Li' + 0 = 0$$

$$q' = i$$

- Write down the ODE for the LRC circuit with no voltage source and $R = 10.0$, $C = 0.5$, and $L = 0.5$. Fill in the matrix A that make the ODE the a first order system $\frac{d}{dt} \begin{pmatrix} q \\ i \end{pmatrix} = A \begin{pmatrix} q \\ i \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. What would be a suitable time interval to plot over?

$$A = \begin{pmatrix} 0 & 1 \\ -\frac{1}{0.5^2} & -\frac{10}{0.5} \end{pmatrix}$$

$$\begin{pmatrix} q \\ i \end{pmatrix}' = \begin{pmatrix} q' \\ i' \end{pmatrix} = \begin{pmatrix} i \\ -\frac{1}{L}q - \frac{R}{L}i \end{pmatrix} = \begin{pmatrix} 0q + 1i \\ -\frac{1}{L}q - \frac{R}{L}i \end{pmatrix}$$

$$i' = -\frac{R}{L}i - \frac{1}{L}q$$

$$\begin{pmatrix} q \\ i \end{pmatrix}' = \begin{bmatrix} 0 & 1 \\ -\frac{1}{L} & -\frac{R}{L} \end{bmatrix} \begin{pmatrix} q \\ i \end{pmatrix}$$

2. Write down the ODE for the LRC circuit with no voltage source and $R = 100.0$, $C = 0.1$, and $L = 0.1$. Fill in the matrix A that make the ODE the a first order system $\frac{d}{dt} \begin{pmatrix} q \\ i \end{pmatrix} = A \begin{pmatrix} q \\ i \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. What would be a suitable time interval to plot over?

$$A = \begin{pmatrix} & \\ & \end{pmatrix}$$